

SQL Mini Project for Beginners

Basic SQL Syntax

Select data

```
select column_name from table_name;
```

Filter data

```
select * from table_name where condition;
```

Grouping & aggregation

```
select column, sum(column)
from table_name
group by column;
```

Filtering grouped data

```
select column, sum(column)
from table_name
group by column
having sum(column) > value;
```

Sorting results

```
select column
from table_name
order by column desc;
```

Dataset

Create table

```
create table sales (
product_id int,
product_name varchar(255),
sale_date date,
quantity int,
revenue decimal(10,2)
);
```

Insert data

```
insert into sales (product_id, sale_date, product_name, quantity, revenue) values
(1, '2021-01-01', 'Product Ab', 50, 100.00),
(2, '2021-01-01', 'Product Ab', 20, 1080.00),
(3, '2021-01-04', 'Product An', 10, 199.00),
(4, '2021-01-04', 'Product Ab', 90, 900.00),
(5, '2021-01-05', 'Product An', 19, 399.00),
(6, '2021-01-05', 'Product Akk', 100, 599.00),
(7, '2021-01-01', 'Product Akk', 99, 4900.00);
```

Practice Questions

1. Retrieve all data from the table

2. Show only records for Product An
3. Find total revenue per day
4. Find total revenue per product
5. Find top-selling products by quantity
6. Which products were sold on January 4th?
7. What is total revenue for Product Ab?
8. Which days had total revenue greater than 150?
9. Which products had total quantity > 30?
10. What is average revenue for Product Akk?

Answer Key

1.

```
select * from sales;
```

2.

```
select * from sales where product_name = 'Product An';
```

3.

```
select sale_date, sum(revenue) from sales group by sale_date order by sale_date;
```

4.

```
select product_name, sum(revenue) from sales group by product_name;
```

5.

```
select product_name, sum(quantity) from sales group by product_name order by sum(quantity) desc;
```

6.

```
select product_name from sales where sale_date = '2021-01-04';
```

7.

```
select sum(revenue) from sales where product_name = 'Product Ab';
```

8.

```
select sale_date, sum(revenue) from sales group by sale_date having sum(revenue) > 150;
```

9.

```
select product_name, sum(quantity) from sales group by product_name having sum(quantity) > 30;
```

10.

```
select avg(revenue) from sales where product_name = 'Product Akk';
```